

Parkinson's disease and cannabinoids: a narrative review of use, nabilone sleep effects, and receptor biology

Abstract

Parkinson's disease (PD) is a synucleinopathy in which nigrostriatal dopamine loss produces bradykinesia, rigidity, tremor, and gait failure, while sleep disruption, pain, anxiety, psychosis, and skin change add much of the lived burden. Levodopa remains the most effective motor replacement, yet it does not restore the circuits that fail around dopamine, and it does not license a cannabinoid as disease-modifying therapy. People with PD already use plant cannabis or would accept cannabidiol (CBD) at high rates, while physicians who manage PD remain far more cautious than other clinicians. That mismatch is not proof of motor efficacy. The only placebo-controlled human contrast in this literature that survives as a sleep signal is a synthetic tetrahydrocannabinol (THC) analogue, nabilone, in an enriched non-motor sample; motor scores in the same programme do not move. Short oral CBD, with or without a small THC fraction, has not shown a dermatologic seborrheic benefit in a nested randomised analysis, and clinic syntheses of motor trials report non-significant pooled change on examiner-rated motor scales. Imaging of cannabinoid receptor type 1 (CB1) availability tracks postural instability, rigidity, and memory rather than tremor, which is a different claim from occupancy by CBD. Toxin and α -synuclein models still show viability, tyrosine hydroxylase, and phosphorylated α -synuclein signals for CBD, a cannabigerol quinone, and a CB2 ligand; those assays are not human motor approval. The practical implication is to keep non-motor cannabinoid pharmacology and motor disease-modification as separate questions rather than one plant-based hope.

Keywords

Parkinson's disease; cannabidiol; nabilone; CB1 receptor PET; non-motor symptoms; α -synuclein models

Key Summary Points

- A substantial minority of people with PD already use cannabis for symptoms, and large web samples report high acceptability of CBD and of higher-THC cannabis; these are attitude and prevalence data, not randomised efficacy.

- Nabilone improves night-time sleep items under placebo-controlled withdrawal and in an open-label six-month extension, without a motor MDS-UPDRS II/III signal; the evidence grade is one enriched RCT post-hoc plus one unblinded extension.
- Oral CBD-rich extract at 2.5 mg/kg/day for about 16 days does not change seborrheic dermatitis scores versus placebo; parent motor results of that programme are not in the dermatology report.
- Lower CB1 PET availability associates with worse postural-instability/gait, rigidity, episodic memory, and executive tests; tremor and bradykinesia do not show that association.
- Cellular and animal models still report neuroprotection-like readouts for CBD and related ligands; those findings remain preclinical.

1. Introduction

Parkinson's disease (PD) is a progressive synucleinopathy whose motor diagnosis still rests on slowness, rigidity, rest tremor, and postural instability, while the biology that kills cells is not confined to those signs. Lewy pathology, α -synuclein aggregation, and nigrostriatal denervation concentrate in the substantia nigra pars compacta, the midbrain nucleus that supplies dopamine to the striatum. When those neurons fail, striatal medium spiny neurons lose the dopamine they need to gate corticostriatal and thalamocortical loops, and the basal ganglia stop selecting movement with the usual balance of direct-pathway facilitation and indirect-pathway braking. Bradykinesia and rigidity follow that circuit failure more closely than rest tremor does; gait and postural instability often worsen later and respond less completely to dopamine replacement. The same disease also produces a long list of non-motor symptoms: fragmented night-time sleep, insomnia, REM sleep behaviour disorder, pain, anxiety, depression, constipation, orthostatic hypotension, cognitive slowing, psychosis, and seborrheic dermatitis. Those complaints are not decorative. They drive quality of life, caregiver load, and many clinic visits, and they are only partly explained by nigrostriatal dopamine loss.

The dopaminergic lesion remains the practical core of treatment. Levodopa, converted to dopamine in remaining terminals and in other decarboxylase-containing cells, still produces the largest examiner-rated motor gains of any oral drug. Dopamine agonists, monoamine oxidase-B inhibitors, and catechol-O-methyltransferase inhibitors extend or spare levodopa. Deep brain stimulation of the subthalamic nucleus or globus pallidus internus can restore motor selection when fluctuations and dyskinesia become unmanageable. None of those tools restores the neurons that died, and none reliably treats the full non-motor list. Long-term levodopa brings motor fluctuations and levodopa-induced dyskinesia, in which the same dopamine pulse that restores walking can produce involuntary choreiform or dystonic

movement. Orthostatic hypotension, daytime sleepiness, impulse-control disorders, and hallucinations constrain agonist use. Sleep and pain often persist after the motor examination looks better. A person with PD can therefore be “optimised” on milligrams of levodopa equivalent and still lie awake, hurt, or fear falling. That residual burden is the clinical soil in which cannabis products grow, with or without a licence.

Cannabis sativa is not one medicine. Δ^9 -tetrahydrocannabinol (THC) is a partial agonist at cannabinoid receptor type 1 (CB1) and cannabinoid receptor type 2 (CB2), the receptors that basal-ganglia physiology reviews place on corticostriatal terminals, medium spiny neurons, and inducible immune cells rather than on the dopaminergic soma itself [1]. It is the principal intoxicating constituent of high-THC flower and of several licensed synthetics. Cannabidiol (CBD) binds CB1 with low affinity, acts as a negative allosteric modulator at that receptor in many assays, and engages TRPV1, 5-HT1A, PPARs, and inflammatory transcription programmes that do not require CB1 occupancy [2]. Plant oils and smoked or vaporised flower mix THC, CBD, cannabigerol, terpenes, and unidentified minor constituents in ratios that a clinic cannot reconstruct from a patient’s report of “cannabis.” Licensed synthetics are a different class again. Nabilone is a synthetic THC analogue and CB1/CB2 partial agonist, historically used for chemotherapy-induced nausea. Dronabinol is synthetic THC. Nabiximols is an oromucosal THC:CBD spray licensed for multiple-sclerosis spasticity in some jurisdictions. Purified CBD (Epidyolex/Epidiolex) is licensed for specified childhood epilepsies, not for PD. Clinic reviews written for movement-disorder practice therefore warn against treating smoked flower, milligram-range CBD isolate, and sub-milligram nabilone as interchangeable interventions [3,4].

The receptors those ligands meet are not interchangeable either. CB1 is among the most abundant G-protein-coupled receptors in the brain and is densely expressed in striatum, globus pallidus, and substantia nigra pars reticulata, sitting on corticostriatal glutamatergic terminals and on striatal medium spiny neurons rather than on the dopaminergic cell bodies themselves [1,5]. Endocannabinoids—chiefly anandamide and 2-arachidonoylglycerol, hydrolysed by fatty acid amide hydrolase and monoacylglycerol lipase—act as retrograde messengers that can dampen glutamate or GABA release depending on the synapse. CB2 is sparser in healthy neurons and more readily induced on microglia and peripheral immune cells, which is why a CB2 ligand in an α -synuclein model is not a CB1 PET finding in disguise [1]. Both CB1 agonists and CB1 antagonists have been reported to modify levodopa-induced dyskinesia in animals, a bidirectional pattern that already cautions against a single “cannabinoid for PD” slogan [1]. Hafida and colleagues summarise a related staging idea from the older biochemical literature: CB1 may fall early after dopamine depletion and then, with CB2 and endocannabinoid tone, rearrange as the lesion matures [2]. That rearrangement is a pathophysiological map, not a dosing table.

Human use has outrun that map. Surveys in Europe find that people with PD already take cannabis among other natural products for symptoms, and that acceptability of CBD and of higher-THC cannabis is high even where THC is not a licensed PD indication [4]. German neurology updates cite self-use prevalences from 8.3% to 37% across questionnaires, with many users stopping for lack of effect, while formal cannabinoid prescriptions for PD remain a tiny fraction of national totals [4]. Patient-reported tremor benefit in large convenience samples sits beside meta-analyses of randomised motor trials that do not show a convincing examiner-rated motor effect [3,4]. The disagreement is not mysterious once endpoints are named: uncontrolled flower, heterogeneous THC:CBD ratios, placebo expectancy, and a tremor item on a questionnaire are not the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) Part III.

Two scientific stories therefore run in parallel, and they should not be forced into one. The first is a non-motor pharmacology story: a defined CB1/CB2 partial agonist, nabilone, can change sleep and pain items in people with PD who already have a heavy non-motor load, without moving motor Parts II and III. The second is a disease-modification story: CBD and related ligands still look protective in 6-hydroxydopamine, reserpine, MPTP, and viral α -synuclein models, while short clinical CBD courses have not delivered a motor or even a dermatologic win in the randomised reports discussed below. Imaging adds a third layer rather than a tie-breaker. Lower CB1 availability on [18F]FMPEP-d2 positron emission tomography tracks postural instability, gait, rigidity, episodic memory, and executive set-shifting, not tremor. Psychosis-related gray-matter loss aligns spatially with CB1 gene expression from a healthy postmortem atlas, not with CB2. Those maps say the endocannabinoid system is in the disease. They do not say that giving CBD, which occupies CB1 poorly, will repair the map.

The aim of this review is to compare what people with PD already use and will accept, what nabilone and oral CBD actually change in clinical protocols, how CB1 appears in living PD and in psychosis maps, and why animal models still look more encouraging than clinics, without pooling milligrams of nabilone with milligrams per kilogram of CBD or treating a garbled survey as a trial.

2. Methods

2.1 Search and sources

Records were identified from a Scopus bibliographic export of journal articles and reviews on PD, parkinsonism, synucleinopathy, and cannabinoid neurology. Screening used titles and metadata first, then public full texts only (publisher PDF or Europe PMC JATS). Paywalled files were not opened. Two additional public full texts were retrieved after the literature table

to fill named interpretation gaps: a post-hoc sleep analysis of the nabilone NMS-Nab trial and a 2022 basal-ganglia endocannabinoid physiology review. A short-term CBD plus THC PD randomised trial identified in the export (and its cognitive-safety companion) was not available as a public full text and is therefore not quoted for motor or fluency numbers.

2.2 Eligibility

Eligible original papers asked a cannabinoid, cannabis, CBD, THC, nabilone, or CB1/CB2 question in PD, a PD toxin or synuclein model, or PD-related human biology (PET, psychosis maps). Eligible reviews discussed cannabinoids or CBD in PD or in neurology with an explicit PD section. Full texts without a cannabinoid question (imaging or biomarker neighbours, mixed non-cannabinoid cocktails, psychedelic or antipsychotic-target papers) were excluded after reading. Reviews were kept in a separate catalog from original articles so that secondary summaries are not treated as new experiments.

2.3 Study selection

After title exclusion, retrieval failure, and full-text exclusion, 31 full texts were included: 19 original reports and 10 reviews from the export that remained eligible, plus the two additional papers named above. Extraction used structured notes (question, design, n, intervention, primary result, limits). Findings were not meta-analysed. Designs and endpoints are incommensurable; the article reports them by theme rather than as a pooled effect.

3. What people with PD already use and will accept

How often people with PD already take cannabis for symptoms depends on who is asked, where they live, and whether the denominator is cannabis or any “natural” product. In the Dutch population-based PRIME-NL cohort, a four-item survey nested among 566 invited participants yielded 367 respondents (mean age 70.3 years, mean disease duration 7.1 years) who were asked only about products used to relieve PD symptoms, not about recreation [6]. Thirty-six percent (132 of 367) had ever used at least one listed or “other” natural health product; cannabis was the second most named product after coffee, at 13.1% (48 of 367), with 4.9% reporting both [6]. A conservative sensitivity analysis that treated all 199 non-responders as non-users lowered any-product prevalence to 23% (132 of 566) [6]. Natural-product use associated with younger age (odds ratio 0.97, 95% CI 0.94–0.99) and not with gender, education, disease duration, or Hoehn and Yahr stage [6]. German neurology updates cite a wider self-use band of 8.3% to 37% across questionnaires, with many users stopping for lack of effect [4]. Those bounds look like disagreement until the instrument is named: PRIME-NL is a clinic-linked Dutch cohort with a four-item list and a 65% response, and its cannabis-specific 13.1% sits inside Hidding’s 8.3–37% band rather than contradicting it [4,6].

The 36% figure is not a cannabis-only prevalence. Convenience clinic samples sit higher because they count users, not the PD population: a 261-user sample used cannabis for tremor in 66% and had stopped in the prior six months in 23%; a Norwegian sample of 530 emphasised sleep (52.5%) and pain (37%) as perceived benefits [3]. Those percentages cannot be averaged with PRIME-NL's 13.1%, because one denominator is people with PD and the other is people already using cannabis [3,6]. Among Dutch users, 39% were aware of possible interaction with PD medication and 39% had discussed use with a neurologist or PD nurse, so even a modest 13.1% is already a disclosure problem [6].

Acceptability of a hypothetical prescription is a different question from documented use, and it splits by chair in the same country. The French CANNABAPA web survey of 1136 adults with self-reported PD (median age 68 years, median time since diagnosis 7 years; 89.1% reached via France Parkinson) found cannabis acceptability of 81.7% (moderate 49.3%, absolute 32.4%; low 18.3%) and CBD acceptability of 87.4% (moderate 45.0%, absolute 42.4%; low 12.6%), with absolute acceptability higher for CBD (z-test $p < 0.001$) [7]. Moderate or absolute acceptability associated with GAD-2 scores of 3 or more, with better cannabinoid knowledge, and with having informed oneself about medical cannabis, and it fell when perceived dependence risk was high or very high (cannabis adjusted odds ratio 0.16, 95% CI 0.09–0.28, $p < 0.001$; CBD 0.23, 0.12–0.44, $p < 0.001$) [7]. Among 218 French health professionals who reported managing PD (45 physicians, of whom 42 were neurologists; 173 non-physicians), cannabis acceptability was low in 27.5% (60/218) and high in 22.9%; CBD acceptability was low in 11.9% and high in 51.4% [8]. Being a physician reduced the odds of moderate or high acceptability by 87% for cannabis (adjusted odds ratio 0.13, 95% CI 0.06–0.28, $p < 0.001$) and by 90% for CBD (0.10, 0.04–0.27, $p < 0.001$) [8]. Low cannabis acceptability was 57.8% in physicians versus 19.7% in non-physicians; a very serious perceived dependence risk further reduced cannabis acceptability (adjusted odds ratio 0.26, 0.08–0.88, $p = 0.031$) [8]. The two Barré surveys therefore describe the same French health system from opposite chairs: patients say they would use CBD even more readily than higher-THC cannabis, while physicians, especially neurologists in a small subsample, discount that willingness because they do not see effectiveness evidence [7,8]. France Parkinson oversampling and a web format likely inflate patient information-seeking; the provider sample has only 45 physicians and an unknown response rate [7,8]. Most respondents still supported easing medical-cannabis legal restrictions in some form, including 82.2% of physicians, so the split is about evidence, occupation, and perceived harm rather than prohibition [8]. Dutch ever-use of 13.1% and French hypothetical CBD acceptability of 87.4% can both be true because one is behaviour in a clinic cohort and the other is stated willingness in an association-enriched web sample [6,7].

Survey benefit reports and examiner-rated motor syntheses look like a fight until the endpoint is named. de Freitas and Fox summarise a 1,373-person sample in which higher Δ9-THC products were more often called beneficial for tremor than similar THC/CBD products (odds ratio 1.7, 95% CI 1.17–2.57, $p = 0.01$), with little subjective benefit claimed for other motor problems including dyskinesia [3]. Two meta-analyses (15 studies including 6 randomised trials; 18 studies including 5 randomised trials) found no convincing motor-score benefit of heterogeneous cannabis preparations [4]. Uncontrolled flower, heterogeneous THC:CBD ratios, placebo expectancy, and a tremor item on a questionnaire are not the Movement Disorder Society Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) Part III [3,4]. That is why clinics hear “it helps my tremor” and still cannot write a motor indication. Those sentences are not new outcome data.

One paper in the sample must be read as a methods warning rather than clustered with PRIME-NL or CANNABAPA. Caiza Asitimbay and co-authors label an original article on cannabis in PD and multiple sclerosis, tabulate $n = 450$ in a “PD” group within $n = 900$, and report pre–post Likert changes after “three weeks of cannabis-based therapy” with ANCOVA p -values [9]. The methods mix a survey frame with a pre–post therapeutic frame, do not specify dose, product, THC:CBD ratio, or diagnostic verification, and include a sentence about “cloud-based healthcare adoption” that does not match the stated aims [9]. The tabulated PD age bands (130 respondents aged 18–35 years and 320 aged 36–55 years, with no ≥ 56 band) are incompatible with typical late-life PD epidemiology and with the Dutch and French samples above [6,7,9]. Linear-regression “odds ratios” are presented for Likert outcomes that are not clinical motor endpoints [9]. The authors themselves state that causality cannot be inferred and that the work should guide future trials rather than conclude effectiveness [9]. This review therefore does not use those ANCOVA F statistics as evidence that cannabis improves PD. Table 1 records the paper so that the warning is visible, not so that $n = 450$ can be pooled with PRIME-NL.

Table 1. Survey and acceptability studies of cannabis or CBD among people with PD and clinicians (Caiza Asitimbay 2025 is included only as a methods warning).

Study	Design	Population (n)	Endpoint	Result
Diadhiou 2024 [6]	Cross-sectional 4-item survey in PRIME-NL	People with PD, Netherlands ($n = 367$ of 566 invited)	Ever-use of cannabis for PD symptoms among listed natural products	36% any listed product (132/367); cannabis 13.1% (48/367); coffee 15.8%
Barré 2024 [7]	Nationwide online acceptability survey	Self-reported PD, France ($n = 1136$)	Moderate/absolute acceptability of cannabis vs CBD	Cannabis 81.7%; CBD 87.4%; absolute

				CBD > cannabis, $p < 0.001$
Barré 2026 [8]	Online provider survey	PD-managing clinicians, France (n = 218; 45 physicians)	Moderate/high vs low acceptability	Physician aOR 0.13 cannabis and 0.10 CBD vs others, both $p < 0.001$
Caiza Asitimbay 2025 [9]	Unclear mixed survey / pre-post; internally inconsistent	Tabulated “PD” subgroup n = 450 of 900	Likert emotional, cognitive, fatigue, “physiological weight” ANCOVA	Not used as efficacy; implausible PD age mix; product/dose unstated

Taken together, Table 1 supports a social fact and forbids a therapeutic leap. A Dutch clinic cohort already contains a 13.1% cannabis-for-symptoms minority [6]. French patients say they would accept CBD more than higher-THC cannabis, and both at high rates [7]. French physicians, especially neurologists in a small subsample, discount that acceptability because they do not see effectiveness evidence [8]. Hidding’s range of 8.3–37% self-use and frequent discontinuation for lack of effect is consistent with that tension rather than contradictory of it [4]. What this heading cannot show is whether the 48 Dutch users improved, which product they took, or whether a physician who became more accepting would then see MDS-UPDRS change. Those questions require defined products and controlled follow-up, which is the next section.

4. Clinical pharmacology in PD: nabilone, CBD, and mixed oils

The only placebo-controlled human cannabinoid contrast among the retrieved clinical papers that isolates a non-motor domain with numeric withdrawal data is nabilone, a synthetic THC analogue, in the NMS-Nab programme, and the six-month extension asks whether that signal persists in the same people without a placebo arm. Sleep was not the parent trial’s inclusion criterion. A post-hoc analysis restricted the 38 randomised participants to the 31 who had MDS-UPDRS item 1.7 (night-time sleep problems) of 2 or more at baseline [10]. During open-label nabilone, item 1.7 and Non-Motor Symptoms Scale Domain 2 (sleep/fatigue) improved ($p < 0.001$); 24 of 31 (77.4%) no longer had a relevant sleep problem (item $1.7 \leq 1$) at randomisation, and 30 of 31 (96.8%) improved by at least one point [10]. In the double-blind withdrawal, placebo worsened MDS-UPDRS-1.7 by 2.00 points (95% CI 1.32–2.68, $p = 0.004$, effect size 1.70) while continued nabilone changed it by 0.12 points ($p = 1.000$); the between-group contrast was $p < 0.001$ [10]. Placebo NMSS Domain 2 worsened 9.57 points (3.24–15.90, $p = 0.004$) versus nabilone, with a between-group difference of 8.57 (2.24–14.91); insomnia (NMSS question 5) drove much of that worsening, whereas daytime sleepiness on the Epworth scale and MDS-UPDRS-1.8 was not the main driver [10]. Twenty-two prior responders who had completed double-blind treatment without a qualifying drug-

related serious, moderate, or severe adverse event entered the open-label extension; 21 entered the analysis set and 19 completed six months, at a mean daily nabilone of 0.87 ± 0.44 mg (median 0.75 mg) after titration [11]. Mean levodopa-equivalent dose did not change significantly by visit 3 (667 ± 521 mg, $p = 0.943$ versus screening) [11]. Clinical Global Impression of Improvement showed non-motor improvement in 78.90% at visit 3 (CGI visit 1 versus visit 3 $p = 0.002$) [11]. NMSS Domain 2 fell 8.26 points (95% CI -13.82 to -2.71 , $p = 0.004$, effect size -0.72); MDS-UPDRS-1.7 fell 1.42 points (-2.16 to -0.68 , $p = 0.002$, effect size -0.92); King's Parkinson's Disease Pain Scale total fell 8.00 points (-15.05 to -0.95 , $p = 0.046$, effect size -0.55); MDS-UPDRS-1.9 (pain) fell 0.74 points (-1.21 to -0.26 , $p = 0.008$) [11]. MDS-UPDRS Part I total change was -2.58 ($p = 0.052$). Parts II and III and a motor sum were all $p > 0.513$ [11]. Mini-Mental State Examination ($p = 0.298$) and Montreal Cognitive Assessment ($p = 0.775$) did not change from parent-trial screening to visit 3 [11]. Those two papers are the same programme, the same ligand, and largely the same responders: enriched-enrolment randomised withdrawal versus unblinded continuation [10,11]. The dissociation that pooling "cannabinoids for PD" would erase is already inside that programme: night-time sleep and pain items move at less than 1 mg/day, while motor Parts II and III do not [11]. The extension cannot prove six-month efficacy versus placebo; the sleep analysis is post-hoc, has no polysomnography, and has no CBD comparator [10,11]. Concentration difficulties were the most common treatment-related adverse event (possibly related $n = 2$); five serious adverse events in four patients were rated unrelated; one participant left for night-time sleep problems requiring prohibited brotizolam [11]. Former nabilone versus former placebo assignment in the parent trial did not split extension efficacy (all $p > 0.427$) [11]. What this programme can show is product-specific: a CB1/CB2 partial agonist at sub-milligram doses can change sleep items in an enriched PD sample without a motor win.

Oral CBD in this sample is a different molecule, a different dose class, and a different endpoint from nabilone, which is why a seborrheic null cannot falsify Peball's sleep items and cannot be averaged with them. Weber and colleagues nested a dermatologic secondary analysis in a parallel, double-blind, placebo-controlled trial of sesame-oil CBD-rich extract (100 mg/mL CBD and 3.3 mg/mL THC) dosed at 2.5 mg/kg/day CBD for PD motor symptoms [12]. Participants took 1.25 mg/kg each morning for 4 ± 1 days, then twice daily for 10 ± 4 days, about 16 days in total; 53 adults with idiopathic PD provided paired facial photographs (26 CBD, 27 placebo) scored on SEDASI (0–60) [12]. Mean final CBD plasma level used in generalised estimating equations was 49.29 ng/mL [12]. Seborrheic dermatitis was present in 12/26 (46.2%) before CBD and 9/26 (34.6%) after ($p = .26$), versus 15/27 (55.6%) to 18/27 (66.7%) on placebo ($p = .26$); the treatment-effect p-value for presence was .12 [12]. The post versus pre risk ratio for seborrheic dermatitis was 0.69 (95% CI 0.41–1.18, $p = .15$) on CBD and 1.20 (0.88–1.65, $p = .26$) on placebo; groups differed at $p = .07$ [12].

Mean SEDASI went from 3.63 to 3.79 on CBD ($p = .81$) and from 5.39 to 4.65 on placebo ($p = .35$); the treatment effect was $p = .38$ [12]. Baseline severity was low; exposure was short because of drug availability; the parent trial was powered for motor symptoms, which this paper does not report [12]. Microgram-range THC:CBD oil is a third clinical object. Ruver-Martins and colleagues randomised six Brazilian adults with moderate–severe idiopathic PD (Hoehn and Yahr III–IV, MDS-UPDRS III >20 in best ON, stable levodopa) 1:1 to oral extract at THC:CBD 250:28 $\mu\text{g/day}$ or 1000:112 $\mu\text{g/day}$ for 90 days, without a placebo arm [13]. HPLC of the extract showed THC $8.7\% \pm 0.5 \text{ m/m}$ and CBD $0.75\% \pm 0.15$ [13]. Insomnia Severity Index improved significantly only in the 1000:112 $\mu\text{g/day}$ group after 60 days ($p = 0.03$ versus baseline, Friedman test with Dunn post-hoc) [13]. Montreal Cognitive Assessment increased more in the lower-dose group in descriptive plots, without a significant difference as stated by the authors; Epworth Sleepiness Scale did not change in either group [13]. Randomising six people between two doses does not create a controlled trial. The $p = 0.03$ insomnia signal is a within-group contrast at one time point in $n = 3$, not a replication of Peball’s withdrawal design, and the milligram gap between 0.87 mg nabilone and 1000 μg THC is three orders of magnitude in the wrong direction for naïve pooling [11,13]. Weber’s 49.29 ng/mL is a short oral CBD course aimed at skin, not a nabilone occupancy experiment, and SEDASI is not MDS-UPDRS-1.7 [10,12]. Head-to-head nabilone versus CBD for insomnia does not exist here. Disease modification in humans does not exist here.

Clinic-facing reviews, which are secondary and add no new n , already ranked this pattern and should be read as the field’s own clustering, not as extra patients. de Freitas and Fox report a meta-analysis of three randomised trials with a small non-significant UPDRS III difference versus placebo; two small open-label inhaled-cannabis studies with UPDRS III change within 30 minutes; open-label CBD with UPDRS III change versus baseline over 2–4 weeks; three small randomised trials without levodopa-induced dyskinesia improvement on CBD/THC products; nabilone with a small single-dose dyskinesia signal in seven people and no significant motor change over four weeks in 38; and nabilone 0.25 mg daily to 1 mg twice daily with efficacy on the UPDRS non-motor subscale, particularly sleep, insomnia, and fatigue ($n = 38$) [3]. CBD 75–300 mg for 14 weeks in REM sleep behaviour disorder improved sleep satisfaction without a difference versus placebo in nights with dream-enacting behaviour [3]. A cognitive randomised comparison of high-dose CBD 100 mg plus low THC 3.3 mg ($n = 29$ as cited) showed worse verbal fluency and concentration versus placebo and adverse events at least twice placebo [3]. Hidding and colleagues reach the same motor conclusion from two larger meta-analytic umbrellas and add that CBD did not benefit PD REM sleep behaviour disorder in Almeida 2021 or restless legs in a 2023 post hoc, while nabilone’s anxiety and sleep effects in NMS-Nab are the PD signal they rate as positive [4]. Hafida and colleagues, restricting attention to CBD, count seven clinical trials in their abstract, four of which concluded no improvement in UPDRS motor severity; Carroll 2004

Cannador (n = 17) showed no McGill pain benefit; a depression subset was negative; sleep results mixed a four-patient open-label RBD anecdote against later negative randomised sleep primaries; anxiety reduction on a simulated public-speaking test after 300 mg CBD (n = 24) is the named randomised non-motor CBD win [2]. Fowler and colleagues state the motor-cognitive line more bluntly still: acute CBD at liver-safe doses generally does not appear effective for PD motor and cognitive symptoms, with Chagas 2014 (n = 7 on 300 mg/day) showing no UPDRS benefit but PDQ-39 quality-of-life improvement, and with open-label high-dose Epidiolex motor change offset by cholestatic enzyme rises in 5 of 13 patients that resolved after stopping [14]. Białoń and colleagues, in a 107-study neurology CBD review, likewise place PD among indications where preclinical work looks better than clinical motor or cognitive superiority versus placebo [15]. Šolea and colleagues still list Lotan's 22-person smoked-cannabis 30-minute UPDRS change and Leehey's open-label Epidiolex titration (total UPDRS improved 17.8%, motor subscale 24.7% in that uncontrolled series; 10 of 13 completed) as PD rows, then conclude that core motor evidence is limited and inconsistent [16]. Those reviews agree on ranking: nabilone non-motor yes, in a small enriched programme; CBD motor no, on the randomised evidence they trust; smoked or open-label signals remain uncontrolled. They are not additional trials. The pharmacological lesson is receptor- and product-specific: a CB1/CB2 partial agonist at sub-milligram doses can sedate the night-time item list in an enriched PD sample; a low-affinity CBD course aimed at skin, and by extension the short oral CBD tradition in PD, has not produced a positive primary in the randomised files that were opened.

Table 2. Clinical cannabinoid exposure in PD with numeric endpoints available in the included full texts.

Study	Design	Population (n)	Endpoint	Result
Peball 2022 [10]	Post-hoc of EERW RCT (nabilone vs placebo withdrawal)	PD with MDS-UPDRS-1.7 ≥ 2 (n = 31 of 38 randomised)	MDS-UPDRS-1.7 in double-blind phase	Placebo worsen 2.00 (1.32–2.68), p = 0.004; nabilone 0.12 (ns); between-groups p < 0.001; OL 77.4% no relevant sleep problem
Peball 2024 [11]	Open-label 6-month extension; mean 0.87 mg/day	Prior nabilone responders (n = 22 enrolled / 19 completers)	NMSS sleep/fatigue; night-time sleep; KPPS; MDS-UPDRS II/III	NMSS Domain 2 –8.26, p = 0.004; item 1.7 –1.42, p = 0.002; KPPS –8.00, p = 0.046; II/III all p > 0.513
Weber 2024 [12]	Nested secondary analysis of parallel RCT	Idiopathic PD (n = 26 CBD / 27 placebo); ~16 days 2.5 mg/kg/day CBD	SEDASI; SD presence RR	SEDASI treatment p = 0.38; SD RR 0.69, p = .15
			ISI at 60 days	

Ruver-Martins 2024 [13]	Open-label two-dose case series (no placebo)	H&Y III–IV PD (n = 6; 3 per dose)		Significant ISI improvement only at THC:CBD 1000:112 µg/day, p = 0.03 vs baseline
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Table 2 is the clinical core of this review. Nabilone’s sleep items move under withdrawal and remain improved without placebo at six months; motor Parts II and III do not [10,11]. CBD-rich extract at a short 2.5 mg/kg/day course does not change SEDASI [12]. A six-person oil series produces one insomnia p-value at a microgram dose that cannot be mapped onto nabilone milligrams [13]. Clinic reviews add the missing motor meta-analytic sentence—non-significant UPDRS III—and the missing RBD sentence—CBD not better than placebo on nights with dream enactment—without this review re-extracting those primary PDFs [2,3,4,14,15,16]. Head-to-head nabilone versus CBD for insomnia does not exist here. Disease modification in humans does not exist here. The pharmacological lesson is receptor- and product-specific: a CB1/CB2 partial agonist at sub-milligram doses can sedate the night-time item list in an enriched PD sample; a low-affinity CBD course aimed at skin, and by extension the short oral CBD tradition in PD, has not produced a positive primary in the randomised files that were opened.

5. CB1 in living PD and psychosis maps

If nabilone is a CB1-competent ligand and CBD is not, the next question is what CB1 looks like in the living PD brain, not whether a plant product occupies it. Two [18F]FMPEP-d2 papers from the same tracer programme ask that question on different endpoint families in overlapping but not identical samples. Ajalin and colleagues measured volume of distribution (VT), an index of CB1 availability, in 15 people with idiopathic PD confirmed by [18F]FDOPA PET (age 66 ± 6 years, disease duration 10 ± 5 years, Hoehn and Yahr 2.5, UPDRS-III OFF 23 ± 8) [17]. Voxel-wise regression of VT against motor subscores, with age and sex as covariates and FDR $p < 0.05$, showed a negative correlation with postural instability and gait disorder ($p = 0.002$; cluster 266,770 voxels across both hemispheres) and with rigidity ($p < 0.001$; cluster 627,855 voxels including thalamus and basal ganglia), whereas tremor was $p = 0.995$, bradykinesia $p = 0.980$, and total UPDRS-III $p = 0.144$ [17]. A CB1–FDOPA correlation did not survive FDR [17]. Findings held under non-parametric tests, region-of-interest analysis, and partial-volume correction [17]. In 23 people with PD (16 men, 7 women; age 65 ± 6 years; disease duration 7.7 ± 4.5 years; UPDRS-III 23.6 ± 8.5 ; levodopa-equivalent 555 ± 274 mg; none using cannabis), voxel-wise regressions with levodopa-equivalent dose as covariate and FDR at voxel $p < 0.01$ linked poorer episodic memory to lower VT [18]. Wechsler Logical Memory I showed two clusters ($p < 0.001$, $T_{\text{peak}} = 5.07$, $kE = 166177$, $q = 0.006$; $p = 0.007$, $T_{\text{peak}} = 4.75$, $kE = 42916$, $q = 0.008$); Logical Memory II showed one cluster ($p < 0.001$, $T_{\text{peak}} = 5.12$, $kE = 288007$, $q = 0.002$);

Trail Making Test B minus A associated with lower VT ($p < 0.001$, $T_{\text{peak}} = 5.11$, $kE = 229035$, $q = 0.004$) [18]. Language, visuospatial function, and working memory showed no relationship [18]. Partial-volume correction essentially replicated the map; winsorising one Trails outlier shrank but did not abolish the executive cluster [18]. The two papers share tracer, disease, and a topographic logic: lower CB1 availability tracks axial/rigid motor features and the memory/executive tests that fail first in many PD clinics, not tremor or bradykinesia [17,18]. They differ in n (15 versus 23), in the covariate set, and in the endpoint (motor subscores versus a neuropsychological battery) [17,18]. PET was acquired on usual medication while some motor scores are listed OFF, so ON/OFF pairing needs care; group cognitive scores did not differ significantly from a small healthy-control column, and only three of 23 met PD mild cognitive impairment, so a dementia contrast was impossible [17,18]. Five of 15 with CSF Alzheimer biomarkers had two or three abnormal values [18]. Cross-sectional VT cannot be said to predict cognitive decline, and neither paper is a nabilone occupancy study [17,18]. Survey patients who name tremor as the reason they use cannabis are therefore pointing at the motor feature least tied to CB1 VT in this PET series [3,17].

Psychosis is a different CB1 question again, and it is not occupancy. Pisani and colleagues took Hedges' g maps of gray-matter volume loss from a prior coordinate-based meta-analysis of 10 voxel-based morphometry studies comparing PD psychosis ($n = 211$) with PD without psychosis ($n = 298$) and correlated those maps with CB1 (CNR1) and CB2 (CNR2) mRNA from six healthy Allen atlas donors [19]. Unadjusted CB1 spatial correlation was $r = 0.337$, $t(76) = 3.122$, uncorrected $p = 0.003$, spatial-autocorrelation-corrected $p < 0.001$; after levodopa-equivalent adjustment, $r = 0.357$, $t(76) = 3.328$, corrected $p = 0.015$ [19]. Unadjusted CB2 was $r = 0.221$, $t(76) = 1.976$, corrected $p = 0.090$; levodopa-adjusted CB2 corrected $p = 0.132$ [19]. Volume and gene-expression data come from different brains; the atlas is not PD tissue; coordinate peaks are coarser than statistical maps; the analysis cannot show that cannabis causes or treats PD psychosis [19]. It can show that the topography of extra gray-matter loss in PD psychosis aligns with where healthy brains express CB1, not CB2, which is congruent with CB1's cortical and basal-ganglia density in the circuit review and incongruent with treating a peripheral CB2 ligand as a psychosis therapy on PET grounds [1,19]. The method is not Ajalin's VT: one cluster is living occupancy correlated with motor and cognitive scores, the other is a spatial correlation of atrophy maps with a healthy postmortem atlas [17,18,19]. Both are compatible with a CB1-rich disease anatomy. Neither licenses CBD occupancy of those voxels.

Table 3. Human CB1 imaging and spatial gene-expression maps in PD.

Study	Design	Sample (n)	Endpoint	Result
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Ajalin 2025 [17]	Cross-sectional [18F]FMPEP-d2 PET vs motor subscores	Idiopathic PD (n = 15)	VT vs PIGD, rigidity, tremor, bradykinesia	PIGD p = 0.002; rigidity p < 0.001; tremor p = 0.995; bradykinesia p = 0.980
Ajalin 2026 [18]	Cross-sectional PET vs neuropsychological battery	Idiopathic PD (n = 23); no cannabis use	VT vs Logical Memory I and TMT B–A	Logical Memory I q = 0.006; TMT B–A q = 0.004; language/visuospatial/ working memory ns
Pisani 2026 [19]	VBM meta-analysis maps vs Allen CB1/CB2 mRNA	211 PDP + 298 PDnP (MRI); 6 donors (mRNA)	Pearson r, spatial- corrected p	CB1 r = 0.337, t(76) = 3.122, corrected p < 0.001; CB2 ns after spatial correction

Table 3 should not be read as a nabilone target-engagement study. None of these participants was dosed with a cannabinoid for the imaging paper [17,18,19]. Lower VT with worse PIGD and memory is compatible with endocannabinoid involvement in axial and cognitive PD, and compatible with why a CB1-competent agonist might change sleep without changing tremor [10,11,17,18]. It is not compatible with the inference that CBD, a low-affinity CB1 ligand, will occupy the voxels that fail on Logical Memory I. It also does not predict who will respond to nabilone; that enrichment experiment has not been done. Wang and colleagues' circuit review remains the vocabulary layer: CB1 on terminals and medium spiny neurons can reshape glutamate and GABA in the same nuclei where dopamine is missing, and CB1 versus CB2 functions diverge [1]. CB2 inverse-agonist data in rats, taken up in the next section, live on the other receptor and should not be averaged with Ajalin's VT [1,17].

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6. Models: why animals still look better than clinics

The preclinical–clinical gap is not a disagreement about a shared number. It is a change of species, dose class, and endpoint. Esfandi and colleagues pretreated PC12 cells with hemp-extracted CBD at 1, 5, or 10 μ M and then applied 100 μ M 6-OHDA for 24 hours [20]. Viability fell to $48.25 \pm 6.65\%$ with toxin versus $97.25 \pm 2.21\%$ in control ($p < 0.0001$, $n = 4$); 5 μ M CBD raised viability to $69.25 \pm 3.50\%$ ($p < 0.001$ versus toxin) and 10 μ M to $81.65 \pm 4.50\%$ ($p < 0.0001$); 5 versus 10 μ M $p = 0.029$ [20]. Apoptosis rose to $36.82 \pm 5.81\%$ with toxin versus $1.5 \pm 1.29\%$ in control and fell to $18.04 \pm 4.05\%$ and $12.60 \pm 2.07\%$ with 5 and 10 μ M CBD; malondialdehyde and reactive oxygen species fell at 5 and 10 μ M, Nrf2 and Bcl-2 mRNA rose, and Bax and caspase-3 mRNA fell (qPCR $n = 3$) [20]. One μ M did not significantly change malondialdehyde versus toxin ($p = 0.128$) [20]. There are no animals, no nigra, and no CB1/CB2 pharmacology in that dish [20]. In vivo, Lima and colleagues gave 6-month-old male Swiss mice reserpine 0.1 mg/kg subcutaneously and CBD 0.5 mg/kg intraperitoneally, either concomitantly or preventively, with typical analysis $n = 9$ –10 per group from 75 animals started [21]. Preventive CBD reduced catalepsy versus reserpine/saline at initial ($p = 0.029$) and intermediate ($p = 0.05$) phases; concomitant CBD delayed catalepsy onset relative to control but did not differ significantly from reserpine/saline on that contrast [21]. Nigral tyrosine hydroxylase–positive counts were higher with CBD than with reserpine/saline on both schedules (concomitant $U = 19.000$, $p = 0.019$; preventive $U = 21.000$, $p = 0.028$), whereas open-field distance was not rescued: concomitant motor-phase distance remained about 2.5–2.7 m in both reserpine groups versus about 10.5 m in vehicle [21]. Formulation then changes the same ligand again. Lapmanee and colleagues compared oral CBD lipid nanoparticles 20 mg/kg, free CBD 20 mg/kg, and levodopa 10 mg/kg in male Wistar rats with high-fat diet, streptozotocin 35 mg/kg, and MPTP 30 mg/kg daily for four weeks, $n = 8$ per group (histology 3–4), with encapsulation efficiency $98.78 \pm 0.90\%$ and particle size 156.33 ± 1.24 nm [22]. CBD-LNP and levodopa decreased catalepsy-bar latency versus diabetic-parkinsonian vehicle; CBD-LNP differed from free CBD on at least one behaviour; striatal dopamine ELISA rose with the nanoformulation more than with free CBD,

though absolute concentrations were not printed [22]. Combined streptozotocin, high-fat diet, and MPTP confounds metabolic rescue with nigrostriatal rescue, and there is no plasma CBD pharmacokinetic table [22]. Those three experiments share CBD and a protection-like readout. They differ in system (PC12 versus male Swiss mouse versus comorbid Wistar rat), in exposure (10 μ M in a dish versus 0.5 mg/kg versus 20 mg/kg oral), and in what “worked” (MTT viability versus TH+ counts without open-field rescue versus bar latency plus dopamine ELISA) [20,21,22]. Fowler’s translational warning is exactly that gap: in-vitro micromolar potencies versus clinical plasma after oral CBD [14]. Weber’s mean 49.29 ng/mL is not 10 μ M, and SEDASI is not TH cell count [12,14,20]. A TH photomicrograph is not a motor trial [21]. Białoń cite Lapmanee as nanoparticle CBD restoring dopamine and behaviour comparably to levodopa in rats; that citation is formulation biology, not a human diabetic-PD trial [15,22].

Viral α -synuclein models move closer to Lewy biology and farther from phytocannabinoid monotherapy, and they should be clustered with each other rather than with Esfandi’s dish. Burgaz and colleagues overexpressed A53T α -synuclein with unilateral nigral AAV9 and gave oral VCE-003.2 20 mg/kg, a cannabigerol quinone derivative described as a PPAR- γ partial agonist without CB1/CB2 activity, for 14 days [23]. Cylinder ipsilateral-paw preference after the vector was prevented ($F(2,17) = 11.50$, $p < 0.001$); vehicle AAV mice lost about 35% nigral TH immunoreactivity ($F(2,21) = 20.67$, $p < 0.0001$); VCE-003.2 restored TH intensity toward sham (Tukey $p < 0.005$ versus vehicle AAV); microgliosis and astrogliosis fell; elevated-body-swing laterality was not rescued ($F(2,19) = 1.49$, not significant) [23]. Joers and colleagues targeted CB2 with SMM-189, a peripheral inverse agonist/modulator, in rats with nigral AAV2/5 human wild-type α -synuclein [24]. In the low-titer cohort designed for inflammation without planned nigral death, daily 6 mg/kg intraperitoneally reduced nigral phosphorylated Ser129 α -synuclein versus vehicle ($t(23) = 3.353$, $p = 0.0023$); proteinase-K-resistant α -synuclein also fell ($p = 0.044$); Western pSer129 fell ($t(15) = 3.710$, $p = 0.0021$) while total human α -synuclein did not ($p = 0.2133$) [24]. DAT showed only a trend ($p = 0.051$) [24]. Those two papers share a compressed viral synucleinopathy and a protection-like readout. They differ in species (mouse versus rat), in ligand (PPAR- γ -active CBG quinone without CB1/CB2 activity versus a CB2 inverse agonist), and in the endpoint that moved (cylinder laterality and TH versus pSer129 optical density without planned cell loss) [23,24]. Neither is nabilone, neither is CBD, and placing SMM-189 next to Ajalin’s CB1 PET is a receptor error the basal-ganglia review already forbids [1,17,24]. Group sizes after outlier removal in Burgaz were ranges (sham 4–6; lesion 8–10), males only, and PPAR- γ dependence in that experiment is inferred from prior toxin work rather than a co-administered antagonist [23].

Not every CB1-positive result in a toxin model is a cannabis product, and striatal CB1 can still matter for calcium biology without anyone dosing a cannabinoid in the animal. Kim and colleagues injected SU-Eohyeol pharmacopuncture, a multi-herb plus deer-antler injectate, at GB34 in male C57BL/6N mice after four MPTP doses of 20 mg/kg, with $n = 10$ per group for von Frey testing [25]. Mechanical hypersensitivity lasted more than 21 days; GB34 injectate reduced pain versus MPTP-vehicle while a gluteal control point and amitriptyline 10 mg/kg did not; CB1 antagonism with SR141716A 3 mg/kg or PPAR γ antagonism with T0070907 5 mg/kg eliminated the analgesic and TH-recovery effects [25]. Rotarod latency improved with the injectate in one results section; the discussion also states that motor deficits were not significantly improved, an internal tension in that paper [25]. Rivas-Santisteban and colleagues showed that CB1 and angiotensin AT1 receptors form heteromers: BRET2 saturation in HEK cells, cross-antagonism on cAMP and ERK, and, in primary striatal neurons, ACEA cutting the Ang II calcium peak by approximately half and returning fluorescence to baseline in less than 90 seconds [26]. In 6-OHDA rats with >95% striatal TH loss, the fraction of NeuN+ neurons with AT1–CB1 heteromers fell from >95% in non-lesioned striatum to about 45% after lesion [26]. Sixteen lesioned rats received levodopa; the extracted text then gives conflicting percentages (about 23% versus about 70%) for dyskinetic versus non-dyskinetic heteromer load, so those recovery numbers are not used as precise efficacy endpoints here [26]. HEK calcium experiments used $n = 5$; proximity-ligation n was at least 5 per group; ACEA was applied in cells, not as a therapeutic in the rats [26]. Kim and Rivas share CB1 as a mechanism, not as a phytocannabinoid prescription: one is an acupoint injectate whose analgesia requires CB1 and PPAR γ , the other is a heteromer assay in which no cannabinoid was given to the lesioned animal [25,26]. Bidirectional CB1 pharmacology in levodopa-induced dyskinesia, already flagged in the circuit review, has a molecular neighbour in these heteromers [1,26].

Docking is not a model of PD at all and should not be clustered with viability or TH. Jamshaid and colleagues docked Cannabis sativa phytochemicals to MAO-B (PDB 2C65) and reported cannabicyclol at -10.8 kcal/mol versus deprenyl/selegiline -7.3 and rasagiline -7.9 kcal/mol, with cannabidiol at -8.4 and cannabigerol at -9.1 [27]. A 100 ns molecular-dynamics run of the cannabicyclol complex showed ligand RMSD drifting to about 2.8 Å after 23 ns; redocking RMSD of the co-crystal was 1.382 Å [27]. There is no enzyme IC₅₀, no cell, and no animal [27]. The authors call for wet assays [27]. This review records the paper so that a computational screen is not silently dropped, and so that cannabicyclol is not confused with CBD in a clinic conversation or averaged with Esfandi's 10 μ M viability [20,27].

Blebea and colleagues' 2025 preclinical phytocannabinoid review is the secondary chorus for this section, not a new n : chronic CBD 10 mg/kg in 6-OHDA reduced nigrostriatal

degeneration via astrocytic TRPV1 and CNTF in the Giuliano work they cite; intrapallidal CBD and GPR55; CBD 100 mg/kg in MPTP with cytokine and NLRP3 changes; Δ^9 -THCV 2 mg/kg delaying levodopa-induced dyskinesia in Pitx3ak mice; VCE-004.8 preserving TH⁺ cells [5]. Kaszyńska adds *C. elegans* CBD against reserpine and α -synuclein locomotion and a hypothesized CBD plus vitamin B12 combination that has no PD trial [28]. Those reviews correctly describe why animals still look better: higher milligram-per-kilogram doses, toxin timing that clinics cannot copy, endpoints (viability, TH, cytokines) that MDS-UPDRS III does not measure, and almost no randomised human disease-modification attempt [2,5,14,15]. The disagreement with Weber, Hafida, Hidding, and Fowler is therefore not a fight about honesty. It is a fight about species, dose, and endpoint [2,4,12,14,20,21].

7. How contemporaneous reviews framed the same question

Contemporaneous reviews, read as secondary literature, already sorted PD cannabinoids into the same bins this article uses. They should not be mined as if they generated new patients. de Freitas and Fox wrote the clinic script: do not endorse cannabis for motor PD; consider insomnia, anxiety, and pain after other options fail; warn about orthostatic hypotension, psychosis, cognition, and smoking [3]. Hidding and Buhmann wrote the German neurology update: meta-analyses negative on motor scores; nabilone positive on anxiety and sleep; CBD negative in RBD; self-use common; PD <1% of national cannabinoid prescriptions [4]. Hafida and colleagues wrote the CBD-only PD overview: preclinical generally encouraging, seven clinical trials insufficient, four motor studies without UPDRS severity improvement, sleep mixed, anxiety positive in one speech-task trial [2]. Those three papers are the clinical voice of the period.

Neurology-wide CBD reviews then place PD below epilepsy. Białoń and colleagues included 107 studies after a multi-database search through December 2025, excluded THC-only products at the eligibility step, and concluded that clinical PD trials “do not have such direct positive effects” as the animal literature, with oral CBD bioavailability about 6–13% (about fourfold with a high-fat meal) and CYP3A4/CYP2C19 interaction risk as translational excuses [15]. Şolea and colleagues used a PRISMA-guided search from 2000 through November 2025 and still could not issue a firm PD efficacy conclusion after mixing Lotan smoked cannabis, open-label psychosis CBD (n = 6), 75 versus 300 mg designs, and Epidiolex titration into one PD table [16]. Fowler and colleagues narrowed the lens to microglia and non-psychotropic phytocannabinoids, found only two PD preclinical studies that directly assayed microglia after those ligands, and stated that liver-safe CBD generally fails motor and cognitive endpoints, with open-label high-dose enzyme safety as the counterweight [14]. Hakami and Alshehri mapped 47 completed ClinicalTrials.gov cannabinoid trials in neurology (from 132 cannabinoid interventional records among

neurological trials), noted PD among the most common conditions and nabilone among common interventions, then centred their comparative narrative on CBD versus THC and did not produce a PD pooled UPDRS result [29]. Phase 2 predominated (18 of 47, 38.3%) [29]. That registry map is activity, not effect size [29].

Mechanistic and policy reviews fill the remaining seats without changing the ranking. Blebea and colleagues synthesise the last years of phytocannabinoid PD models and explicitly withhold clinical efficacy [5]. Kaszyńska proposes CBD plus vitamin B12 as a homeodynamics hypothesis, including liposomes, without a PD combination trial [28]. Shoyama links electroacupuncture analgesia to CB2 in inflammatory-pain models and asserts cannabinoid benefit in PD by citing prior systematic reviews, without reproducing their numbers and without a PD randomised trial of acupuncture plus CBD [30]. de Aquino and colleagues screened 150 records, tabulated 55, discussed 42, and framed *Cannabis sativa* for Brazilian health-system integration; PD appears as one neurological indication with mixed cited clinical significance and animal p-values imported from other papers [31]. Those p-values are not this review's statistics [31]. Wang and colleagues, included as a circuit map rather than as a clinic trial, remain the receptor grammar: altering endocannabinoid tone can have anti-movement-disorder effects, but CB1 and CB2 diverge, and both agonists and antagonists have modified dyskinesia in models [1]. Treating that 2022 physiology review as a new experiment would be a category error [1].

The reviews agree more than their titles suggest. Epilepsy holds the only strong licensed CBD claim; PD holds surveys, small nabilone non-motor data, and a preclinical–clinical gap [2,3,4,5,14,15,16,29]. Where they differ is emphasis: clinic papers lead with “do not use for motor symptoms”; microglia papers lead with NLRP3 and then admit the clinical failure; policy papers lead with regulation. None of them claims that pooling nabilone milligrams with CBD mg/kg and smoked flower is valid. None treats Caiza Asitimbay as a PD RCT. None equates CB1 PET VT with a CBD occupancy target. Those are the forced themes this narrative refuses for the same reasons.

Discussion

Two readings of the same sample remain plausible, and they are not equally supported at the same level of design. The reading that fits the randomised human data is a non-motor cannabinoid story: a synthetic THC analogue, nabilone, improves night-time sleep items when placebo withdrawal is the contrast and continues to improve sleep and pain items over six open-label months, at a mean 0.87 mg/day, without moving MDS-UPDRS II or III [10,11]. Clinic reviews had already placed that signal on anxiety and sleep and had already declined a motor indication [3,4]. The reading that fits the animal data is a neuroprotection story: CBD at

10 μ M or 0.5–20 mg/kg, a CBG quinone at 20 mg/kg, and a CB2 inverse agonist at 6 mg/kg change viability, TH, cylinder laterality, or pSer129 α -synuclein [20,21,22,23,24]. Those assays have not been matched by short oral CBD in PD on SEDASI, nor by the UPDRS III meta-analyses the clinic reviews cite [2,3,4,12,14]. Preferring the first reading for practice and the second for mechanism is not inconsistency. It is matching claim to design.

Pooling fails for ordinary pharmacological reasons, not because the papers “disagree” about a shared number. Nabilone is a CB1/CB2 partial agonist at sub-milligram oral doses [11]. CBD is a low-affinity, multi-target molecule given in mg/kg in Weber and in micrograms of THC:CBD in Ruver-Martins [12,13]. Smoked flower in Lotan, as cited by Şolea, is a 30-minute uncontrolled motor snapshot [16]. Peball’s primary sleep items are MDS-UPDRS-1.7 and NMSS Domain 2 [10,11]. Weber’s primary in the opened paper is SEDASI [12]. Ajalin’s endpoints are PET VT correlations [17,18]. Esfandi’s endpoint is MTT viability [20]. Joers’ endpoint is pSer129 optical density [24]. There is no common scale, no common ligand, and no common species. A random-effects diamond drawn across those rows would be a statistical fiction.

Survey benefit and RCT motor nulls can both be true. Thirteen percent of a Dutch PD cohort used cannabis for symptoms [6]. French patients accept CBD (87.4%) more than cannabis (81.7%) [7]. Physicians have 87–90% lower odds of finding those products acceptable [8]. de Freitas’ 1,373-person tremor odds ratio for higher THC is a subjective report [3]. Hidding’s meta-analyses are examiner-rated motor scores [4]. Uncontrolled use, placebo, product heterogeneity, and the difference between a tremor complaint and Part III explain the split without requiring anyone to be lying [3,4,6,7,8]. Caiza Asitimbay cannot be used to close that split; its internal inconsistencies and implausible PD age structure remove it from the efficacy column [9].

CB1 imaging and CB2 models are not a controversy. Lower CB1 VT with worse PIGD, rigidity, Logical Memory I ($q = 0.006$), and Trails B–A ($q = 0.004$), plus CB1 mRNA alignment with psychosis-related atrophy ($r = 0.337$, spatial-corrected $p < 0.001$), describe a CB1-rich disease anatomy [17,18,19]. SMM-189 lowering pSer129 ($t(23) = 3.353$, $p = 0.0023$) describes a CB2-immune intervention in rats [24]. Wang’s review already states that the two receptors diverge in the same basal ganglia [1]. CBD’s clinical failure at liver-safe doses does not refute Esfandi’s 10 μ M dish, because Weber’s plasma nanograms and Esfandi’s micromoles are not the same exposure, and because SEDASI is not TH cell count [12,14,20]. Preventive CBD’s TH preservation without open-field rescue in reserpine mice is the same lesson inside one paper [21].

Limitations of the evidence are those of the studies. Nabilone sleep data are a post-hoc subset of an enriched withdrawal trial without polysomnography [10]. The extension has 19

completers and no placebo [11]. Weber is underpowered for seborrheic dermatitis, lasts about 16 days, and withholds parent motor results [12]. Ruver-Martins is $n = 6$ [13]. PET samples are $n = 15$ and $n = 23$ [17,18]. Most animal papers use males only and toxin or compressed AAV phenotypes [21,22,23,24,25,26]. Reviews mix smoked cannabis into CBD tables [16]. A registry systematic review does not extract PD effect sizes [29]. Human disease modification, head-to-head nabilone versus CBD for insomnia, and CB1 PET as a nabilone-response biomarker are unmeasured [1,10,17]. Those are scientific holes.

Clinical meaning follows the same calibration. A movement-disorder clinician can treat nabilone as a possible non-motor option for sleep and pain in selected patients who resemble NMS-Nab responders, with attention to cognition, orthostasis, and confusion, and without promising a motor gain [3,4,10,11]. The same clinician should not tell a patient that plant cannabis is an evidence-based treatment for tremor or that CBD at wellness-shop doses will protect the nigra [3,4,12,14]. Patient acceptability is a counselling fact: people will ask, some already use, and many will not tell the neurologist [6,7,8]. Disclosure, interaction awareness (only 39% of Dutch natural-product users), and product unknownness are the practical risks [6]. Imaging can inform pathophysiology conversations; it cannot yet stratify cannabinoid therapy [17,18,19]. Preclinical programmes on VCE-003.2, SMM-189, and CBD nanoparticles remain at the stage of asking whether PPAR- γ , CB2 immune tone, or formulation—not “cannabis”—is the variable [22,23,24].

Conclusions

PD still needs dopamine replacement, and it still leaves sleep, pain, and other non-motor symptoms on the table after levodopa is optimised. People with PD already use cannabis for symptoms and say they would accept CBD even more readily than higher-THC cannabis, while physicians remain sceptical for lack of effectiveness evidence. Nabilone, a defined CB1/CB2 partial agonist, is the only product in this literature with placebo-controlled worsening of night-time sleep items on withdrawal and with persistent sleep and pain-item change at six months, without motor Parts II and III moving. Short oral CBD has not shown a seborrheic benefit in a nested randomised analysis, and clinic syntheses of motor randomised trials remain non-significant on examiner-rated scales. CB1 availability in living PD tracks gait, rigidity, and memory rather than tremor, and psychosis-related atrophy tracks CB1 gene-expression maps rather than CB2; those maps do not license CBD occupancy. Toxin and α -synuclein models continue to show protection-like readouts that clinics have not reproduced on motor scales, at doses and endpoints that do not match. Next measurements that would actually decide the two stories are a pre-specified nabilone versus CBD insomnia trial with polysomnography, a CB1 PET enrichment study for nabilone response, and a long, defined-ratio motor trial on MDS-UPDRS III. Until those exist, the honest clinical sentence is narrow:

nabilone may help selected non-motor sleep and pain items; cannabis is not an evidence-based PD motor therapy; animal hope is still animal hope.

References

- [1] Wang M, Liu H, Ma Z. Roles of the cannabinoid system in the basal ganglia in Parkinson's disease. *Front Cell Neurosci*. 2022;16:832854.
- [2] Hafida EG, Rachid S, Halima G, Najib K. CBD's potential impact on Parkinson's disease: an updated overview. *Open Med (Wars)*. 2024;19(1):20241075.
- [3] de Freitas MET, Fox SH. Advice to people with Parkinson's in my clinic: cannabis. *J Parkinsons Dis*. 2024;14(4):873-881.
- [4] Hidding U, Mainka T, Buhmann C. Therapeutic use of medical cannabis in neurological diseases: a clinical update. *J Neural Transm (Vienna)*. 2024;131(2):117-126.
- [5] Blebea NM, Pușcașu C, Hancu G, Stăniș AM, Chiriță C. Recent preclinical evidence on phytocannabinoids in neurodegenerative disorders: a focus on Parkinson's and Alzheimer's disease. *Pharmaceuticals (Basel)*. 2025;18(6):890.
- [6] Diadhiou S, Maas BR, Schootemeijer S, Bloem BR, de Vries NM, Calon F, Darweesh SKL, de Rus Jacquet A. Natural health products for symptomatic relief of Parkinson's disease: prevalence, interest, and awareness. *J Parkinsons Dis*. 2024;14:1257-1264.
- [7] Barré T, Cazorla G, Di Beo V, Lopez F, Radoszycki L, Maradan G, Baunez C, Carrieri P. Acceptability of and attitudes to the therapeutic use of cannabis and cannabidiol in people with Parkinson's disease: a French survey. *Clin Park Relat Disord*. 2024;11:100286.
- [8] Barré T, Cazorla G, Di Beo V, Bianchi C, Baunez C, Carrieri P. Healthcare providers' acceptability of cannabis and cannabidiol to manage Parkinson's disease in France. *Curr Ther Res Clin Exp*. 2026;104:100830.
- [9] Caiza Asitimbay MA, Kapoor T, Maqsood R, Reddy P, Sahoo LK, Aggarwal P, Lyngdoh NNK. An examination of the application of cannabis in therapeutic use for Parkinson's syndrome. *Health Leadersh Qual Life*. 2025;4:772.
- [10] Peball M, Seppi K, Krismer F, et al. Effects of nabilone on sleep outcomes in patients with Parkinson's disease: a post-hoc analysis of the NMS-Nab trial. *Mov Disord Clin Pract*. 2022. doi:10.1002/mdc3.13471.

- [11] Peball M, Heim B, Carbone F, Schorr O, Werkmann M, Ellmerer P, Marini K, Krismer F, Knaus HG, Poewe W, Djamshidian A, Seppi K. Long-term safety and efficacy of open-label nabilone on sleep and pain in Parkinson's disease. *NPJ Parkinsons Dis.* 2024;10:61.
- [12] Weber I, Zagana-Prizio C, Sivesind TE, Adelman M, Szeto MD, Liu Y, Sillau SH, Bainbridge J, Klawitter J, Sempio C, Dunnick CA, Leehey MA, Dellavalle RP. Oral cannabidiol for seborrheic dermatitis in patients with Parkinson disease: randomized clinical trial. *JMIR Dermatol.* 2024;7:e49965.
- [13] Ruver-Martins AC, Martinez IAR, Holla VG, Souza BL, Silva EG, Novoa DMA, Souza-Silva E, Araujo FS, Prediger RD, Nascimento FP. Low doses of cannabis extract ameliorate non-motor symptoms of Parkinson's disease patients: a case series. *Front Hum Neurosci.* 2024;18:1466438.
- [14] Fowler DK, Savage TM, Mackie DI. Cannabidiol and other non-psychotropic cannabinoids from *Cannabis sativa* as therapeutics for microglial-mediated neuroinflammation and neurodegeneration. *J Cannabis Res.* 2026;8:82.
- [15] Białoń M, Kędziora M, Starowicz K. Cannabidiol in neurology: current insights and translational perspectives. *Pharmaceuticals (Basel).* 2026;19(2):330.
- [16] Şolea R, Şerban E, Călugăr-Solea SF, Mathe E, Blebea NM, Hancu G, Serban G. Cannabis and cannabinoids: the medical potential of cannabidiol in mental and neurological disorders. *Pharmaceuticals (Basel).* 2026;19(8):1238.
- [17] Ajalin R, Al-Abdulrasul H, Tuisku J, Hirvonen J, Lahdenpohja S, Rinne JO, Brück A. Impaired gait, postural instability, and rigidity in relation to CB1 receptor availability in Parkinson's disease. *Mov Disord.* 2025;40(1):163-167.
- [18] Ajalin RM, Al-Abdulrasul H, Tuisku JM, Karrasch M, Hirvonen J, Kirjavainen AK, Zetterberg H, Rinne JO, Brück AE. Deficits in episodic memory and executive functions in relation to CB1 receptor availability in Parkinson's disease. *J Parkinsons Dis.* 2026. doi: 10.1177/1877718X261457830.
- [19] Pisani S, Váša F, Velayudhan L, Bhattacharyya S. Gray matter volume loss in Parkinson's disease psychosis and cannabinoid receptor gene expression in the brain. *Mov Disord.* 2026;41(6):1566-1571.
- [20] Esfandi A, Mehrafarin A, Kalateh Jari S, Naghdi Badi H, Larijani K. Cannabidiol extracted from *Cannabis sativa* L. plant shows neuroprotective impacts against 6-HODA-

induced neurotoxicity via Nrf2 signal transduction pathway. *Iran J Pharm Res.* 2025;24(1):e160499.

[21] Lima AC, Bioni VS, Becegato MS, Meier Y, Cunha DMG, Aguiar NA, Gonçalves N, Peres FF, Zuardi AW, Hallak JEC, Crippa JA, Smaili SS, Abilio VC, Silva RH. Preventive beneficial effects of cannabidiol in a reserpine-induced progressive model of parkinsonism. *Front Pharmacol.* 2025;16:1539783.

[22] Lapmanee S, Bhubhanil S, Wongchitrat P, Charoenphon N, Inchan A, Ngernsutivorakul T, Dechbumroong P, Khongkow M, Namdee K. Assessing the safety and therapeutic efficacy of cannabidiol lipid nanoparticles in alleviating metabolic and memory impairments and hippocampal histopathological changes in diabetic Parkinson's rats. *Pharmaceutics.* 2024;16(4):514.

[23] Burgaz S, Navarro E, Rodríguez-Carreiro S, Navarrete C, Garrido-Rodríguez M, Lastres-Becker I, Chocarro J, Lanciego JL, Muñoz E, Fernández-Ruiz J. Investigation in the cannabigerol derivative VCE-003.2 as a disease-modifying agent in a mouse model of experimental synucleinopathy. *Behav Brain Funct.* 2024;20:28.

[24] Joers V, Murray BC, McLaughlin C, Oliver D, Staley HE, Coronado J, Achat-Mendes C, Golshani S, Kelly SD, Goodson M, Lee D, Manfredsson FP, Moore BM II, Tansey MG. Modulation of cannabinoid receptor 2 alters neuroinflammation and reduces formation of alpha-synuclein aggregates in a rat model of nigral synucleinopathy. *J Neuroinflammation.* 2024;21:240.

[25] Kim JI, Yeo H, Jo Y, Kim H, Park SM, Kim NS. SU-Eohyeol pharmacopuncture ameliorates Parkinson's disease-associated pain via the CB1 and PPAR γ pathways in an MPTP-induced mouse model. *Pain Res Manag.* 2026;2026:3334432.

[26] Rivas-Santisteban R, Muñoz A, Lillo J, Raïch I, Rodríguez-Pérez AI, Navarro G, Labandeira-García JL, Franco R. Cannabinoid regulation of angiotensin II-induced calcium signaling in striatal neurons. *NPJ Parkinsons Dis.* 2024;10:220.

[27] Jamshaid A, Aziz M, Raza U, Mazhar R, Shakil MAU, Sajid M, Tang L, Sehgal SA. Exploring the therapeutic potential of cannabis constituents in Parkinson's disease: insights from molecular docking simulations. *Medinformatics.* 2026;3(3):260-274.

[28] Kaszyńska AA. Cannabinoids: potential for modulation and enhancement when combined with vitamin B12 in case of neurodegenerative disorders. *Pharmaceutics (Basel).* 2024;17(6):813.

[29] Hakami AY, Alshehri FS. Therapeutic potential of cannabinoids in neurological conditions: a systematic review of clinical trials. *Front Pharmacol*. 2025;16:1521792.

[30] Shoyama Y. Therapeutic potential of the cannabinoid system and acupuncture in neurodegenerative brain disorders: a comprehensive review. *Innov Acupunct Med*. 2025;18:11.

[31] de Aquino PEA, Xavier JMC, Fonteles MMF, Leal LKAM, Viana GSB. Therapeutic potential and challenges of Cannabis sativa in neurological disorders: pathways to integration into health systems. *Rev Bras Plantas Med*. 2025;27:e2025002.